

ENTERPRISE MOBILE ARCHITECTURE

Web to Android Migration Master Plan

Complete 30-Day Transition from React Web Application to Native Android App
Built with React Native & Expo SDK 51 · Zero Backend Disruption

30

DAYS TOTAL

4

PHASES

60+

FEATURES

6

ROLES

8

MODULES

WEEK 1 · DAYS 1-7

Foundation & Setup

- Android Studio & Expo SDK
- JWT Auth & SecureStore
- Role-Based Routing (6 roles)

WEEK 2 · DAYS 8-14

Core Hospital Modules

- Receptionist Dashboard
- Doctor EMR, Vitals & Prescriptions
- Lab Tests & Appointments

WEEK 3 · DAYS 15-21

Billing & Admin Suite

- Razorpay Payments & UPI
- Invoices & Financial Reports
- Patient Portal & Notifications

WEEK 4 · DAYS 22-30

Polish, Test & Launch

- Offline Cache & Biometric Auth
- 6-Role Testing Matrix
- Production APK & Google Play

TECHNOLOGY STACK ALIGNMENT

React Native

Expo SDK 51

Expo Router

Zustand

React Query

NativeWind

Axios

Razorpay RN

Victory Charts

expo-print

EAS Build

 **Core Project Objective:** Transition the Swastik Hospital Management System from a browser-dependent web portal into a fast, native Android app for staff tablets and smartphones. All 60+ clinical and administrative features will remain 100% synchronized with the live hospital database.

**About Swastik Hospital Platform**

Swastik HMS is a specialized psychiatric hospital platform governing all inpatient (IPD) and outpatient (OPD) care. Current operations include emergency admission protocols, rounding logs, ICD-11 psychiatric diagnosis tracking, SOAP clinical documentation, ward bed allocation, diagnostic laboratory workflows, and multi-tier medical billing.

While the existing React 19 web application is fully operational, hospital staff face mobility barriers carrying laptops during ward visits and rounds. The Android migration unlocks bedside access for doctors and field staff.

**Non-Negotiable Architecture Decision**

The FastAPI + MongoDB backend remains 100% UNCHANGED.

The Android application functions purely as a new native client consuming the existing REST API endpoints over secure HTTPS. This eliminates any risk of backend regressions, database migrations, or downtime for ongoing hospital operations during the 30-day mobile buildout.

Shared Live Database: Web portal users and Android mobile users simultaneously interact with the same database records in real time.

Business Impact & Operational Advantages

**Bedside Mobility**

Doctors input vitals and SOAP notes right at the patient's bedside without waiting for PC access.

**Instant Push Alerts**

Stat lab alerts, queue callouts, and appointment reminders ring directly on staff phones.

**Offline Resiliency**

Ward staff review cached records and schedules even during basement or WiFi dropouts.

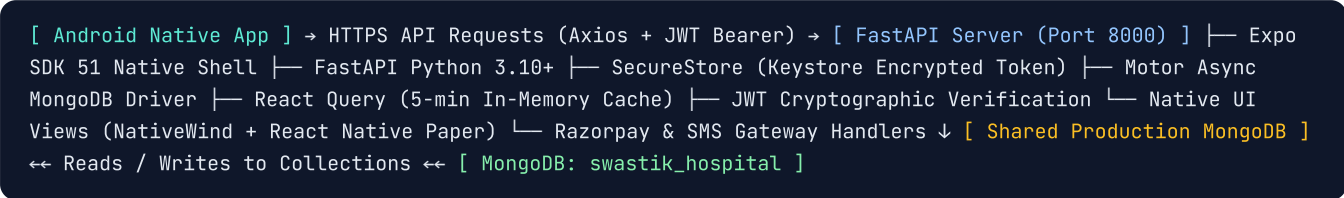
**WhatsApp PDFs**

Prescriptions and lab receipts are shared directly with patients over WhatsApp in one tap.

Project Output & Release Matrix

Deliverable Artifact	Format	Distribution Method	Target Completion
Staff Internal APK	Android .apk	Direct installation link via Google Drive / WhatsApp	End of Week 4 (Day 28)
Production Play Bundle	Android .aab	Google Play Console (Internal Testing & Production Track)	Day 29
Clean Source Code	Git Repository	Private GitLab/GitHub repo with branch protection	Continuous (Day 1-30)
Staff Training Manual	Interactive PDF	Role-specific cheatsheets (Doctor, Reception, Lab, Billing)	Day 30

Client-Server Integration Blueprint



Curated Technology Stack & Industry Rationale

 **React Native + Expo 51**

Shared React logic, zero Java/Kotlin boilerplate, reliable hardware APIs (Camera, Biometrics, Filesystem), and zero build-machine dependency via EAS Cloud.

 **Expo Router + React Nav v6**

File-based routing matching modern Next.js patterns. Native stack back-behavior and custom bottom tab bars for each of the 6 hospital roles.

 **Zustand + React Query v5**

Zustand manages lightweight global auth states. React Query provides automated background re-fetching, cache invalidation, and offline resilience.

 **NativeWind (Tailwind Mobile)**

Enables identical TailwindCSS utility classes from the web app. Eliminates the need to handcraft StyleSheet objects from scratch.

 **react-native-razorpay**

Official Native Android SDK opening embedded UPI apps (GPay, PhonePe, Paytm) and card sheets with cryptographic HMAC verification.

 **Victory Native + expo-print**

Victory Native renders fluid SVG revenue charts. Expo-print compiles HTML strings into vector PDF documents ready for WhatsApp dispatch.

Web to Android Component Translation Guide

Web Architecture (HTML / React 19)	Android Architecture (React Native)	Implementation Behavior
<div> / <section> Layouts	<View> / <SafeAreaView>	Flexbox layout defaulting to flexDirection: 'column'
<p>, , <h1> Typography	<Text>	Mandatory container for all text strings in React Native
Browser localStorage	expo-secure-store	Hardware-backed Android Keystore AES-256 encrypted storage
HTML <table> / Large Lists	<FlatList>	Virtualized windowing rendering only visible viewport items
CSS Modals & Overlays	<Modal> / React Native Paper	Native Android dialog with elevation shadows and backdrop blur
Client-side jsPDF	expo-print + expo-sharing	Renders clean vector PDFs and triggers the Android share sheet

03 — Complete Feature Inventory (60+ Features)

Comprehensive catalog of all modules ported from the web application

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Authentication & Access Control

3 Features

- JWT secure token login with automatic session expiration handling
- Google OAuth 2.0 social login via expo-auth-session
- Dynamic role routing: Directs users instantly to role-specific screens

Receptionist Module

11 Features

- OPD & IPD patient registration with emergency case flagging
- Instant UHID lookup by name, phone number, or date of birth
- Live appointment queue (Confirm, Reschedule, Cancel actions)
- Ward and room allocation visualizer with live bed states
- Inpatient formal admission procedures & consent form capture
- Discharge clearance workflow with summary generation
- Live token ticket counter for doctor consultation queue
- Real-time notifications and emergency alerts

Doctor Clinical EMR Module

15 Features

- Ward Board dashboard tracking active inpatient stability levels
- Bedside EMR suite with tabbed clinical inspection views
- SOAP clinical notes (Subjective, Objective, Assessment, Plan)
- ICD-11 & DSM-5 psychiatric diagnostic code selector
- Medication table with frequency, route, dosage, and duration
- Vertical clinical history timeline with chronological event dots
- Diagnostic lab order dispatch & specimen tracking
- Daily physician rounding log & nursing observation notes
- Bedside vitals logging (BP, Pulse, Temperature, SpO2, Weight)
- Clinical psychiatric risk meter & behavioral flags

Laboratory Diagnostic Module

5 Features

- Pending test order queue with STAT priority highlighting
- Parameterized test result entry with automated reference range validation
- Out-of-range clinical flags highlighted in red
- Official laboratory test report PDF creation & WhatsApp delivery
- Laboratory catalog configuration (Pricing, normal ranges, turnaround)

Medical Billing & Invoicing

9 Features

- Executive revenue dashboard with daily & monthly receipts
- Itemized invoice generation with automated 18% GST calculation
- Native Razorpay gateway (UPI, Credit/Debit cards, Netbanking)
- Dynamic UPI QR code generator for direct cashier counter scans
- Inpatient running bill estimator & discharge clearance bills
- Invoice PDF receipt generation & WhatsApp direct sharing

Patient Self-Service Portal

6 Features

- Self-registration & OTP phone verification
- Doctor appointment booking with real-time calendar slot picking
- Access to personal verified diagnostic laboratory reports
- Complete medical history and prescription PDF downloads

Hospital Administration Control

8 Features

- Executive KPI dashboard (Revenue, Bed occupancy, Patient volume)
- Staff user management (Create, Deactivate, Reset passwords, Roles)
- Departmental configurations & clinical fee schedules
- Financial audit logs & system health status monitors

Mobile-First Enhancements (Exclusive to Android)

 Biometric Fingerprint & Face Unlock

 Push Notifications for Appointments & Results

 Offline Record Caching Engine

 Integrated UHID QR Code Scanner

 Native WhatsApp Document Sharing

 Pull-to-Refresh Gesture on All Views

 60-Second Auto-Sync Background Queries

 System Dark & Light Theme Support



Phase 1: Foundation & Setup

Establish developer tooling, create the Expo TypeScript project, configure the Axios JWT interceptor, build the Zustand auth store, and verify login flows for all 6 roles.

Week 1

DAYS 1-7

Day 1 Environment & Native SDK Setup

Tooling Installation

- 1 Install **Node.js v20 LTS** & **Java JDK 17**. Set `JAVA_HOME` in Windows system variables.
- 2 Install **Android Studio** + Android SDK Platform 34 + Google Play Intel x86 Emulator.
- 3 Configure `ANDROID_HOME` in Environment Variables; verify using command: `adb devices`.
- 4 Install global Expo tooling: `npm install -g expo-cli eas-cli`.

Day 2 Project Scaffolding & Packages

Expo SDK 51 Setup

```
# Initialize Expo Project
npx create-expo-app swastik-mobile -t blank-typescript
cd swastik-mobile

# Install Hardware & Native Packages
npx expo install expo-router expo-secure-store expo-camera \
  expo-notifications expo-local-authentication expo-sharing \
  expo-print expo-image @react-native-picker/picker

# Install State, Networking, UI & Charts
npm install axios zustand @tanstack/react-query nativewind \
  tailwindcss victory-native react-native-svg react-native-razorpay
```

Day 3 Clean Directory Architecture

File Tree Layout

```
swastik-mobile/
├── app/
│   ├── (auth)/login.tsx # Expo Router screens
│   ├── (admin)/         # Login flow
│   ├── (doctor)/        # Admin dashboard & tabs
│   ├── (receptionist)/  # Doctor EMR & ward board
│   ├── (lab)/           # Front desk & admissions
│   ├── (billing)/       # Pathology & test entry
│   ├── (portal)/        # Invoices & Razorpay
│   └── (portal)/home    # Patient self-service
├── services/api.ts      # Axios client with interceptor
├── store/authStore.ts   # Zustand auth state
└── components/         # Reusable atomic UI widgets
```

Days 4-5 API Client & Secure Token Store

```
// services/api.ts
import axios from 'axios';
import * as SecureStore from 'expo-secure-store';

export const api = axios.create({
  baseURL: process.env.EXPO_PUBLIC_API_URL,
  timeout: 25000,
});

api.interceptors.request.use(async (config) => {
  const token = await SecureStore.getItemAsync('token');
  if (token) config.headers.Authorization = `Bearer ${token}`;
  return config;
});
```

Days 6-7 Login Screen & Role Routing

Implement the branded login screen with hospital identity fallback, and automatic routing to role-specific layouts:

```
const ROLE_ROUTES = {
  admin: '/(admin)/dashboard',
  doctor: '/(doctor)/dashboard',
  receptionist: '/(receptionist)/dashboard',
  lab: '/(lab)/dashboard',
  billing: '/(billing)/dashboard',
  patient: '/(portal)/home',
};

// Navigate: router.replace(ROLE_ROUTES[user.role])
```

Phase 1 Verification Checklist

- ☐ Android emulator booted & running Expo SDK 51
- ☐ Backend reachable over HTTPS from mobile device
- ☐ JWT token stored safely in Android Keystore
- ☐ All 6 user roles redirect to their designated views

05 — Phase 2: Receptionist & Doctor Modules (Week 2, Part 1)

Front-desk operations, patient registration, ward board, and bedside EMR

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Phase 2 (Part 1): Receptionist & Doctor Care

Rebuild the core daily clinical workflow: Reception dashboard with live counters, OPD/IPD registration, ward bed tracking, and the doctor's full bedside EMR suite.

Week 2

DAYS 8 – 11

Day 8

Receptionist Live Dashboard

Front Desk Operations

- 1 Implement 4 live stat cards: **Today's Admissions**, **Active Inpatients**, **Available Rooms**, and **Pending Discharges**.
- 2 Port web appointments table into an optimized `FlatList` with patient UHID, doctor name, and time badges.
- 3 Embed action buttons on each appointment card: **Confirm**, **Reschedule**, and **Cancel** triggering native modals.
- 4 Configure React Query with `refetchInterval: 60000` for automatic background queue refreshes.

Day 10

Doctor Ward Board & Patient Queue

Physician Dashboard

- 1 Executive physician overview: Total assigned patients, critical alerts, and discharge candidates.
- 2 Instant typeahead search filtering patients by UHID, full name, or room number with 300ms debounce.
- 3 Color-coded acuity tags: ● Critical Attention, ● Guarded, ● Stable Inpatient.
- 4 Single tap on any patient card smoothly opens the EMR screen via `router.push()`.

Day 9

Patient Registration Engine

OPD / IPD Form Flows

- 1 Wrap form in `KeyboardAvoidingView` to prevent keyboard obstruction on small phone displays.
- 2 Text inputs: Full Name, Guardian, Contact Phone, Government ID, Address, Emergency Contact.
- 3 Native pickers for Gender, Visit Type (OPD/IPD), and Attending Doctor from `GET /api/doctors`.
- 4 Submit to `POST /api/patients/register` and render the auto-generated UHID barcode/QR card.

Day 11

Bedside EMR Suite (5 Clinical Tabs)

Comprehensive Clinical Record

1. **Vitals Tab:** Numeric grid recording BP (Systolic/Diastolic), Pulse, Temp, SpO2, and Weight.
2. **SOAP Notes Tab:** Multiline text areas for Subjective, Objective, Assessment, and Plan observations.
3. **Diagnosis Tab:** Searchable psychiatric ICD-11 & DSM-5 code catalog with tap-to-add tags.
4. **Medications Tab:** Drug name selector, dosage, route (Oral/IV/IM), frequency, and duration.
5. **Clinical Timeline:** Chronological visual timeline of all past notes, tests, and admissions.

✓ Days 8–11 Deliverables Verification

- ☐ Receptionist dashboard auto-refreshes every 60 seconds
- ☐ Patient registration assigns unique UHID successfully
- ☐ Doctor searches and opens EMR for any hospital patient
- ☐ SOAP notes and Vitals commit cleanly to backend database

06 — Phase 2: Prescriptions, Lab & Appointments (Week 2, Part 2)

Prescription PDF dispatch, diagnostic laboratory, and visual calendar scheduling

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Day 12 Prescription Generator & WhatsApp

Medical Documentation

Generate authentic medical prescription PDFs with hospital header, doctor credentials, and medication schedule:

```
import * as Print from 'expo-print';
import * as Sharing from 'expo-sharing';

export async function sharePrescription(rx) {
  const html = `<html><body style="font-family:Arial;padding:20px;">
    <h2 style="color:#0d9488;">Swastik Hospital</h2>
    <p><b>Patient:</b> ${rx.patient_name} | <b>UHID:</b> ${rx.uhid}</p>
    <hr/>
    ${rx.meds.map(m => `<p>${m.name} - ${m.dose} - ${m.freq}</p>`).join('')}
  </body></html>`;
  const { uri } = await Print.printToFileAsync({ html });
  await Sharing.shareAsync(uri, { mimeType: 'application/pdf' });
}
```

Day 13 Diagnostic Lab Queue & Results

Pathology Workflow

- 1 Lab dashboard highlighting pending orders, STAT orders, and completed batches.
- 2 Result entry forms displaying normal clinical reference ranges side-by-side with inputs.
- 3 Automatic alert triggers when values deviate outside normal physiological limits.
- 4 One-tap generation of signed laboratory report PDFs ready for sharing.

Day 14 Interactive Appointment Calendar

Scheduling System

- 1 Integrate react-native-calendars showing month-at-a-glance view with marked appointment dots.
- 2 Tapping any date reveals the itemized appointment queue for that specific day below.
- 3 Reschedule modal allowing selection of alternative consultation dates and timeslots.
- 4 New appointment booking wizard selecting attending doctor, clinic department, and time.

End-to-End Hospital Workflow on Mobile

1. **Patient Arrival:** Reception registers OPD/IPD patient → generates UHID token.
2. **Consultation:** Doctor views queue on mobile Ward Board → opens EMR.
3. **Clinical Documentation:** Doctor logs Vitals & SOAP notes → prescribes meds.
4. **Diagnostic Orders:** Doctor orders lab tests → Lab Tech receives real-time queue item.
5. **Lab Execution:** Lab logs results → generates verified report PDF for patient.
6. **Delivery:** Prescription and lab report sent to patient's phone via WhatsApp.

✓ Phase 2 Complete Milestone Check

- ☐ Doctor creates and exports prescription PDF via WhatsApp
- ☐ Lab orders transition smoothly from Pending to Completed
- ☐ Appointments are confirmed, rescheduled, and booked on calendar



Phase 3: Medical Billing & Razorpay

Implement the financial backbone: executive revenue dashboards, GST itemized invoicing, embedded Razorpay mobile checkout (UPI, cards, netbanking), and payment verification.

Week 3

DAYS 15 – 17

Day
15

Revenue Dashboard & Analytics

Financial
Insights

- 1 Executive metric cards: Today's Collections, Monthly Invoiced, Pending Arrears, and Cashier Balance.
- 2 7-Day revenue trend bar chart rendered natively via Victory Native XL.
- 3 Payment distribution breakdown (UPI vs Cash vs Card vs Insurance TPA).
- 4 Searchable invoice audit trail with status filters (Paid, Unpaid, Partial, Overdue).

Day
16

Itemized Invoicing & GST Engine

Billing
Generation

- 1 Patient typeahead search matching UHID and pre-populating patient billing details.
- 2 Catalog picker adding hospital services (Consultations, Room charges, Tests, Medicines).
- 3 Dynamic tax calculation: Subtotal, 18% Healthcare GST (where applicable), and Grand Total.
- 4 Generates formal hospital invoice PDF ready for export and cashier settlement.

Day 17

Razorpay Payment Integration

Native Payment Flow

Secure Key Separation: The Razorpay Key ID is injected via environment variables. The Key Secret remains strictly on the FastAPI backend for cryptographic signature verification.

```
import RazorpayCheckout from 'react-native-razorpay';

export async function processInvoicePayment(inv) {
  const options = {
    key: process.env.EXPO_PUBLIC_RAZORPAY_KEY_ID,
    amount: inv.total * 100, // Amount in paise
    currency: 'INR',
    name: 'Swastik Hospital',
    description: `Invoice #${inv.id}`,
    prefill: { contact: inv.patient_phone },
    theme: { color: '#0d9488' },
  };

  const paymentData = await RazorpayCheckout.open(options);
  // Cryptographically verify signature on backend
  await api.post('/api/billing/payment/verify', paymentData);
  Alert.alert('Success', 'Payment verified and invoice settled!');
}
```

✓ Days 15–17 Deliverables Verification

- ☐ Revenue graphs display real receipts from production database
- ☐ Invoices compute subtotals, discounts, and GST accurately
- ☐ Razorpay native sheet opens UPI apps (GPay, PhonePe) on Android
- ☐ Backend verifies payment signature and updates invoice to PAID

08 — Phase 3: Admin, Portal & Push Notifications (Week 3, Part 2)

Administrative governance, patient self-service portal, and notification system

Day 18

Hospital Administration Suite

Executive Controls

Full executive management suite across 8 sub-modules:

- **Staff Management:** Create staff logins, assign roles, reset credentials.
- **Departmental Catalog:** Manage clinics, bed tiers, and fee tables.
- **Financial Audits:** Department-wise revenue and expenditure charts.
- **System Health:** Monitor database latency, active sessions, and errors.
- **Audit Logs:** Tamper-resistant audit trails tracking clinical modifications.

Day 19

Patient Self-Service Portal

Patient Mobile Interface

- 1 Patient mobile registration with OTP phone number verification.
- 2 Self-booking engine: Select preferred psychiatrist, date, and morning/afternoon slot.
- 3 Digital health locker: View past laboratory test results with graphical trend lines.
- 4 One-tap download of verified prescriptions and discharge summaries.

Days 20-21

Push Notifications & Ward Grid

Alerts & Bed Occupancy

```
// Device Token Registration
import * as Notifications from 'expo-notifications';

export async function registerDevicePush() {
  const { status } = await Notifications.requestPermissionsAsync();
  if (status === 'granted') {
    const token = (await Notifications.getExpoPushTokenAsync()).data;
    await api.post('/api/notifications/register', { token });
  }
}
```

Automated Push Notifications Triggered for:

- Appointment reminder alerts dispatched 2 hours prior to scheduled consultation.
- Doctor notification when emergency patient arrives at the front desk.
- Patient notification when diagnostic laboratory test results are verified and ready.

Room & Bed Visualizer (Day 21): Color-coded grid showing ● Available Beds, ● Occupied Beds, and ● Maintenance Rooms with single-tap patient reassignment.

☒ Phase 3 Complete Milestone Check

☐ Admin suite modifies staff roles and audits system logs cleanly

☐ Patients self-book appointments and review records on phone

☐ Push notifications successfully deliver to Android notification tray

☐ Ward bed visualizer accurately matches hospital floor state



Phase 4: Quality, Build & Launch

Implement offline caching and biometric unlock, execute the 6-role end-to-end testing matrix, build the test APK for staff evaluation, and submit the production AAB to Google Play.

Week 4

DAYS 22 – 30

Days 22–23

Offline Cache & Biometric Unlock

Hardware Integration

- 1. Offline Cache:** React Query configured with `networkMode: 'offlineFirst'` allows staff to inspect patient files, bed lists, and appointments during network drops.
- 2. Biometric Unlock:** Fingerprint & Face ID unlocks authenticated sessions in under 1 second without password re-entry:

```
import * as LocalAuth from 'expo-local-authentication';

export async function biometricLogin() {
  const result = await LocalAuth.authenticateAsync({
    promptMessage: 'Swastik Hospital Staff Authentication',
    fallbackLabel: 'Use Password',
  });
  return result.success;
}
```

Days 24–26

6-Role End-to-End QA Testing

Quality Assurance Matrix

Admin & Reception:

- Staff creation & permission locks
- OPD/IPD registration & UHID search
- Appointment confirm / reschedule
- Bed allocation & formal discharge

Doctor, Lab & Billing:

- Ward board sorting & EMR tabs
- SOAP notes & Vitals validation
- Lab orders & result ranges
- Razorpay UPI settlement verification

Days 27–29

Production EAS Builds (APK & AAB)

Cloud Build Execution

```
# 1. Login to Expo EAS Cloud
eas login

# 2. Build Testing APK (Day 28)
eas build -p android --profile preview
# Output: SwastikHospital-v1.0.apk (Direct install)

# 3. Build Google Play Production AAB (Day 29)
eas build -p android --profile production
# Output: SwastikHospital-v1.0.aab (Play Store package)
```

☁ **Cloud Compilation:** Builds compile on Expo's dedicated cloud servers in 12–15 minutes, generating clean APK download URLs ready to send to staff via WhatsApp.

Day 30

Hospital Rollout & Staff Onboarding

Go-Live Execution

- 1 Install testing APK on hospital ward tablets and doctors' personal Android phones.
- 2 Host 30-minute hands-on orientation for nursing and front-desk reception teams.
- 3 Monitor live FastAPI server logs for any anomalies or mobile latency bottlenecks.
- 4 🚀 Official transition to mobile hospital operations.

✅ Final Production Readiness Checklist

- ☐ Tested across Android versions 8.0 through 14.0
- ☐ Standalone .apk installs smoothly on staff mobile phones
- ☐ Play Store .aab package signed with release keystore

30-Day Visual Gantt Implementation Roadmap

Days 1–7	Phase 1: Environment, Expo SDK 51, JWT Interceptor, 6-Role Auth & Routing
Days 8–11	Phase 2A: Receptionist Dashboard, OPD/IPD Registration, Doctor Ward Board & EMR
Days 12–14	Phase 2B: Prescription PDFs, WhatsApp Sharing, Lab Tests & Appointment Calendar
Days 15–17	Phase 3A: Medical Billing, Automated GST Invoicing, Razorpay Native UPI Integration
Days 18–21	Phase 3B: Hospital Admin Suite, Patient Portal, Push Alerts, Ward Bed Visualizer
Days 22–30	Phase 4: Offline Engine, Biometrics, 6-Role QA Matrix, APK / Play Store AAB & Launch

Backend REST Endpoints Consumed by Android App



Authentication & User Directory

POST

/api/auth/login

Authenticates credentials & issues JWT token

GET

/api/auth/me

Retrieves logged-in user profile & role

POST

/api/admin/users

Creates new staff credentials (Admin only)



Patients, Admissions & Rooms

GET

/api/patients

Search patients by UHID / Name / Phone

POST

/api/patients/register

Creates new OPD / IPD patient profile

GET

/api/rooms

Retrieves live room & bed occupancy grid

POST

/api/admissions

Formally admits patient to inpatient ward



Clinical EMR & Pathology Lab

GET

/api/emr/{patient_id}

Fetches complete patient EMR records

POST

/api/emr

Saves Vitals, SOAP notes, and Medications

GET

/api/lab/orders

Fetches pending pathology diagnostic queue

PUT

/api/lab/results/{id}

Commits verified test values & generates PDF



Billing & Appointments

GET

/api/appointments

Retrieves appointment schedule for given date

PUT

/api/appointments/{id}

Updates status: Confirm / Reschedule / Cancel

POST

/api/billing/invoices

Generates itemized GST invoice

POST

/api/billing/payment/verify

Verifies Razorpay cryptographic signature

Security & Data Privacy Framework



Hardware-Backed Token Storage:

Session tokens are saved exclusively using expo-secure-store utilizing the Android Keystore system. Tokens cannot be extracted even if the device is plugged into a PC.



Zero Hardcoded Credentials:

API URLs and Razorpay Key IDs are injected via .env files with the EXPO_PUBLIC_ prefix. The .env file is strictly excluded from version control via .gitignore.



Minimal Local PHI Retention:

Patient health information is held temporarily in volatile memory by React Query and purged automatically upon application logout. No medical records are written to unencrypted storage.



Cryptographic Payment Settlement:

Payments are never verified locally on mobile. The mobile app receives Razorpay response tokens and routes them to the FastAPI backend for SHA-256 HMAC signature verification.

Risk Management & Contingencies

Identified Risk	Mitigation Strategy
WiFi dead zones in hospital	React Query offline cache preserves previously fetched records and queue views.
Google Play review delays	Standalone APK distributed directly to staff devices on Day 28, bypassing store approval.
Complex psychiatric EMR	Built with native tabs and specialized pickers matching the web interface identically.

Frequently Asked Questions

Q: Does the website stop working when the Android app is deployed?

A: No. Both the web portal and Android app operate simultaneously connected to the same live FastAPI backend and MongoDB database.

Q: Can doctors use their personal Android phones safely?

A: Yes. All data travels over HTTPS, session tokens are hardware encrypted, and biometric unlock prevents unauthorized device access.

Q: How do we roll out app updates after launch?

A: Using Expo EAS Update, non-native code fixes and UI improvements can be published over-the-air instantly without requiring staff to reinstall APKs.

Executive Project Approval & Sign-off

By signing below, stakeholders approve the 30-day Android Migration Plan, technical scope, architectural boundaries, and resource commitments:

Medical Director

Swastik Hospital

Chief Technology
Lead

Development Team

Operations Head

Project Sponsor